Saussure in HLLSet Algebra

Part IV: Geographical Linguistics

A Computational Exegesis of the *Course in General Linguistics*,
Part Four, in the Boolean Lattice and GF(2) Ring of HLLSet Algebra

Fourth paper in the series *Saussure in HLLSet Algebra*

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Abstract

Part IV of the series re-founds Part Four of the *Course*, *Geographical Linguistics*, as the multi-world theory of one lattice. Parts I–III kept one question out of sight: *where* the changes happen. Saussure's Part Four is that question, and the algebra answers it with the world layer of the EWM programme. The diversity of languages is the family of **worlds** over the one shared plane: hash configurations and codebooks are dialects, the BSS threshold and the LUT are the boundary of a community, and one lattice carries all of them at the cost of one LUT per world, not one lattice per world. The complexity of geographical diversity—dialect continua, overlapping isoglosses—is the graded geometry of the soft key: worlds share nodes exactly but read them differently, and an isogloss is a threshold on a coordinate profile, not a cut. The causes of diversity are re-founded as Saussure re-founded them: *time* is the primary cause, and the algebra makes time exact—two communities with a common ancestor frame diverge by the product of their transport matrices once their pivots stop being shared; substrata are older generations still readable in the commit DAG. The propagation of linguistic waves is the **bridge** $\Phi_{w_1 \to w_2} = \text{ingest} \circ \text{transcribe} \circ \text{materialize}$: deterministic, idempotent on builders, monotone, carrying encodings, not content; a loan is a node whose provenance crosses two worlds, an inheritance is a node whose provenance stays on one DAG path. The dispatcher closes the geography: it is the speaker who chooses which world actualizes the shared langue. The paper ends with the complete dictionary of Part Four and the residual handed to Part V: reconstruction as time travel through old frames.

Keywords: Saussure; geographical linguistics; diversity of languages; dialect; isogloss; substratum; linguistic wave; bridge; world; hash configuration; codebook; dispatcher; HLLSet; Boolean lattice; GF(2) ring; computational exegesis.

1 Introduction

1.1 Where Part IV stands in the series

Part I [5] re-founded the general principles; Part II [6] re-founded synchronic linguistics as the differential theory of value; Part III [7] released time and re-founded diachronic linguistics as the theory of the pivot—sound change as extension and rotation, analogy as the fourth proportional of GF(2), agglutination as the promotion of a view to an original. All three parts assumed, without saying so, that there is *one* frame, *one* community, *one* language. Part Four of the *Course* removes that assumption.

Saussure's Part Four has four chapters: (I) on the diversity of languages; (II) the complexity of geographical diversity; (III) the causes of geographical diversity; (IV) the propagation of

linguistic waves [2, Part IV]. Part IV of this series follows the same order, after a preliminary section that fixes the world layer.

1.2 Method, unchanged

The method is that of Parts I–III [5, 6, 7]: no metaphor, principles as theorems or invariants, ambiguity recorded as data. The translation note of Part I applies unchanged; references to the *Course* are by part, chapter, and section.

1.3 Roadmap

Section 2 fixes the standing definitions and adds the world layer (worlds, interpretations, bridges). Section 3 re-founds the diversity of languages as a family of worlds over one plane. Section 4 re-founds the complexity of geographical diversity as graded overlap and thresholds. Section 5 re-founds the causes of diversity: time as pivots, separation as unsynchronized generations, substrata as old frames. Section 6 re-founds the propagation of linguistic waves as bridges, with loans separated from inheritances by provenance. Section 7 collects the dictionary; Section 8 states what Part IV adds; Section 9 hands the retrospective residual to Part V.

2 Preliminaries: the synchronic and diachronic inheritance, and the world layer

We restate the standing definitions with their labels; proofs and motivation are in [5, 6, 7].

Definition 2.1 (Plane, lattice, ring, IICA, frames, fibers, pivots)**.** *The* ***plane*** *is* $\mathbb{B} = \{0, \dots, 32767\}$*; the* ***lattice*** *is* $\mathcal{B} = \mathcal{P}(\mathbb{B})$ *under* $\cup, \cap, \subseteq$ *and the Boolean ring* $\Delta = \Delta, \cap$*.* $\mu : \mathcal{T} \to \mathcal{B}$ *is an* ***IICA morphism*** *(idempotent, immutable, content-addressed);* $\rho : \mathcal{B} \dashrightarrow \mathcal{C}$ *materializes through the door. A* ***basis frame*** $F = (B_1, \dots, B_k)$ *is a row-echelon GF(2) basis of observed nodes with distinct leading bits, stamped with a generation* γ*; the hard key of* $X \in \langle F \rangle$ *is* $c \in \mathrm{GF}(2)^k$ *with* $X = \bigoplus_{i:c_i=1} B_i$*, the soft key is* $w_i = \mathsf{BSS}_{B_i}(X)$*, and the spill is* $s = |X \setminus \mathrm{cover}(F)|$*. The* ***synchronic fiber*** *is* $\mathsf{Syn}(t) = (\mathcal{B}, F(t), \gamma(t))$*; a* ***pivot*** *is the extension/rotation of Definition 1 of [7]; the* ***transport*** *of a hard key across a pivot is* $c' = c\mathbf{M}$*.*

2.1 The world layer

Part IV adds the objects that the EWM programme built for multi-user representation [9, 12], now read as Saussure's geography.

Definition 2.2 (World; codebook; community)**.** *A* ***world*** *is a triple* $\mathsf{W} = (h, s, \tau)$*: a hash configuration* (h, s) *inducing the measurement morphism* $\mu_{\mathsf{W}} : 2^{\mathcal{T}} \to 2^{\mathbb{B}}$*; a BSS threshold* τ *selecting the thresholded node set* $\mathcal{N}_\tau(\mathsf{W})$*; and an* ***interpretation*** $\iota_{\mathsf{W}} : 2^{\mathbb{B}} \to 2^{\mathcal{T}}$ *that reads a node's active bits back as tokens through the world's LUT [9]. A* ***codebook*** *is the vocabulary-aware encoder that quantizes a host's encodings to tokens before* μ_{W} *[10]. A* ***community*** *is a set of hosts sharing one world and one generation stamp* γ *of a frame: they speak the same dialect when their LUTs agree and their frames are synchronized.*

Proposition 2.3 (Shared representation, separate interpretation)**.** *Two worlds may share the very same node* $X \in \mathcal{B}$ *and yet be distinct: they are distinct exactly when their interpretations differ on some shared node,*

$$\mathsf{W} \neq \mathsf{W}' \quad \Longleftrightarrow \quad \exists\, X \in \mathcal{N}_\tau(\mathsf{W}) \cap \mathcal{N}_{\tau'}(\mathsf{W}') : \iota_{\mathsf{W}}(X) \neq \iota_{\mathsf{W}'}(X).$$

All worlds live on the one lattice $\mathcal{B}$*; what separates them is the LUT, never the bits.*

Proof. This is the representation/interpretation separation of [9]: the bit vector is semantically inert, and all meaning lives in the LUT, which is a different object for every hash configuration. □

Proposition 2.4 (Holographic multi-worlds)**.** *Let Ω be a family of worlds. Then (i)* common representation: *every world is read through the same lattice $\mathcal{B}$, and the representation is not duplicated per world; (ii)* separate interpretation: *each world adds only its LUT and threshold; (iii)* joint closure: *every Boolean expression over nodes drawn from one world stays in that world's subalgebra. The cost of hosting $|\Omega|$ worlds is $|\Omega|$ LUTs, not $|\Omega|$ lattices.*

Proof. Parts (i)–(iii) are Theorem 2 and the holographic principle of [9]. □

Definition 2.5 (Bridge)**.** *The **bridge** from world W_1 to world W_2 is*

$$\Phi_{\mathsf{W}_1 \to \mathsf{W}_2} \;=\; \mathit{ingest}_{\mathsf{W}_2} \circ \mathit{transcribe} \circ \mathit{materialize}_{\mathsf{W}_1},$$

materializing in the source world, transcribing the recovered encodings, and ingesting under the target world's configuration [9].

Proposition 2.6 (Bridge laws)**.** *$\Phi_{\mathsf{W}_1 \to \mathsf{W}_2}$ is deterministic (IICA); idempotent on builders; monotone: $A \subseteq A'$ implies $\Phi(A) \subseteq \Phi(A')$; and in general a strict coarsening: collisions at materialization can only add atoms to the target. The bridge carries encodings, not content.*

Proof. See [9]: materialize, transcribe, and ingest are deterministic; composition preserves IICA; materialization is exact on builders and monotone in the inclusion order. □

3 Chapter I: On the diversity of languages

Saussure opens Part Four with the plain fact of diversity: languages differ from one another in space, as they differ in time [2, Part IV, ch. I]. The diversity is the datum to be explained, not a deviation from a norm.

Reading 3.1 (Diversity is the family of worlds). One plane $\mathbb{B}$ carries a family Ω of worlds. A language is a world together with the frames its community has actually computed: the hash configuration and codebook are its phonology in the broadest sense, the LUT is its lexicon, and the frame is its grammar. Two languages are two worlds over the same plane—two readings of one lattice, in the sense of Proposition 2.3.

Observation 3.2 (Diversity is computed, not given). The lattice does not come divided into languages. What divides it is *use*: which tokens a community ingests, under which configuration, and which generations it shares. Diversity is therefore not a property of the plane but of the family of worlds and their histories. This is Saussure's point that a language is a social fact: the social fact is the shared LUT and the synchronized generation.

Proposition 3.3 (One lattice suffices for all languages)**.** *For any finite or countable family of worlds Ω, the whole family is a subset of the single lattice $\mathcal{B}$; the representation budget stays $|\mathbb{B}|$ bits, shared, and each world adds one LUT and one threshold. Consequently the diversity of languages costs nothing at the representation level: the plane is the common earth, and the worlds are readings of it.*

Proof. Proposition 2.4: the representation is common, the interpretation is per-world, and the cost is a per-world LUT. □

Remark 3.4 (Language and dialect are the same datatype)**.** *Saussure refuses a principled line between language and dialect: the difference is one of degree and of social-political history, not of structure. The algebra agrees: a language and a dialect are both worlds; the only structural differences are the size of the LUT, the generation of the frame, and the threshold τ. There is no type of world that is a "language" and another that is a "dialect".*

4 Chapter II: Geographical diversity: its complexity

Saussure's chapter II complicates the picture: languages do not sit in clean boxes. Dialects shade into one another, and the lines drawn between them—the isoglosses—never coincide over any distance [2, Part IV, ch. II].

Reading 4.1 (Continua and isoglosses). The continuity of dialects is the graded overlap of interpretations: two worlds whose LUTs agree on many tokens and whose frames are close read the same region of the lattice similarly. An isogloss is a threshold on a coordinate profile: the boundary where one soft key falls below τ.

Definition 4.2 (Dialect continuum; isogloss)**.** *A* ***dialect continuum*** *is a family of worlds* $\mathsf{W}_1, \ldots, \mathsf{W}_n$ *such that adjacent worlds' interpretations overlap on their shared nodes with BSS above threshold,*

$$\mathsf{BSS}_{\mathcal{N}(\mathsf{W}_i)}\big(\mathcal{N}(\mathsf{W}_{i+1})\big) \;\geq\; \tau,$$

for each i*. An* ***isogloss*** *for coordinate* B_i *of a frame is the boundary of the set of communities whose soft key* $w_i = \mathsf{BSS}_{B_i}$ *exceeds* τ*: it is a level set of a graded quantity, not a cut.*

Proposition 4.3 (Boundaries are thresholds, not cuts)**.** *For any two adjacent worlds in a continuum, the set of nodes on which they agree is graded by the soft key: agreement is exact on the shared nodes (Proposition 2.3) and partial elsewhere, and the soft key and the spill measure how far apart the worlds are. Consequently there is no point at which a language "stops"; there are only thresholds below which agreement is deemed insufficient. The complexity of geographical diversity is the complexity of a graded boundary, not of a partition.*

Remark 4.4 (Why isoglosses never coincide)**.** *Each isogloss is a level set of a different coordinate of a different frame; different coordinates change at different rates under pivots, and different frames are synchronized with different communities. There is no reason for the level sets to coincide, and in the algebra they generically do not. This is Saussure's observation that the bundle of isoglosses never forms a single line, stated as the generic transversality of level sets.*

5 Chapter III: Causes of geographical diversity

Saussure's chapter III asks why diversity exists and answers: time is the primary cause. Geographical separation is not itself a cause; it is what *allows* divergence to accumulate without being reabsorbed [2, Part IV, ch. III].

Reading 5.1 (Time as pivots; separation as unsynchronized generations). In the algebra, time is the sequence of pivots: every pivot increments a generation and re-coordinates values. Two communities with a common ancestor frame share the same langue exactly as long as they share the same pivots; once their pivots stop being shared—once each community computes its own extensions and rotations—their generations drift apart and their transported coordinates diverge. Separation is not a force; it is the absence of the bridge that would re-synchronize them.

Proposition 5.2 (Divergence is the product of transports)**.** *Let* W_1 *and* W_2 *share an ancestor frame* $F(0)$*, and suppose that after time* 0 *they undergo disjoint sequences of pivots with change-of-basis matrices* $\mathbf{M}_1(1), \ldots, \mathbf{M}_1(t)$ *and* $\mathbf{M}_2(1), \ldots, \mathbf{M}_2(t)$*. Then the value of a shared node* X *at time* t *is*

$$c_1 \;=\; c\,\mathbf{M}_1(1)\cdots\mathbf{M}_1(t), \qquad c_2 \;=\; c\,\mathbf{M}_2(1)\cdots\mathbf{M}_2(t),$$

and the two communities diverge by the coordinate difference $c_1 \Delta c_2$*. The divergence is zero exactly while the two sequences of pivots coincide; from the first unsynchronized pivot onward, generically,* $c_1 \neq c_2$*. Time, not space, is doing the work: the matrices are pure generation increments.*

Proof. Iterate the transport law of Part III [7]. Coordinate difference is XOR in $\mathrm{GF}(2)^k$; the two sequences coincide if and only if every pivot is shared. □

Observation 5.3 (Substrata are old generations). When a wave of new speakers arrives and a community re-coordinates around them, the older frame does not disappear: every generation remains readable in the commit DAG, and every node keeps its old coordinates under the old frame by exact transport. A *substratum* is therefore not a mysterious residue; it is the set of old generations still referenced in the DAG. The algebra keeps substrata as first-class objects: any node can be read against the frame of any earlier generation, with the spill measuring what that old lens cannot see [10].

Remark 5.4 (Saussure's refusal of internal causes)**.** *Saussure denies that geographical diversity has an internal, linguistic cause. The algebra agrees: the pivot is occasioned by residue (Part III), and residue arrives from parole—from outside the system. Geography merely decides which pivots are shared and which are not. The cause of diversity is the same residual that Part III located, allowed to accumulate in separated worlds.*

6 Chapter IV: Propagation of linguistic waves

Saussure's chapter IV describes how innovations spread: a linguistic wave travels across a territory through communication, and it may be stopped by barriers [2, Part IV, ch. IV].

Reading 6.1 (The wave is the bridge). A linguistic wave is the bridge of Definition 2.5. An innovation—a node, or a pivot with its new opposition—crosses from one world to another by being materialized in the source, transcribed into shared encodings, and ingested in the target. Barriers are LUT separation (no shared codebook across which to transcribe) and threshold rejection (the imported node's soft key falls below τ in the target frame).

Proposition 6.2 (Waves propagate without loss on builders)**.** *Let $\Phi = \Phi_{\mathsf{W}_1 \to \mathsf{W}_2}$ be the bridge. Then Φ is deterministic, idempotent on the builders of the source nodes, and monotone; hence an innovation propagates exactly to the extent that its builders survive materialization, and it can only gain collateral atoms from collisions, never lose a builder. A wave is therefore a monotone, convergent spread: it does not weaken with distance, it is* blocked*—by a missing codebook or a threshold—or it arrives whole.*

Proof. The bridge laws of Proposition 2.6, restated geographically. □

Observation 6.3 (Loans and inheritances are separated by provenance). A node inherited from a common ancestor has a provenance path inside one DAG lineage: its recipe names only ancestors of that lineage. A node borrowed across worlds has a provenance path that *crosses* the bridge: its recipe names a source node in W_1 and a re-ingestion in W_2. The algebra therefore separates loans from inheritances by provenance alone, without any external etymological dictionary. A loan is a view whose sources live in two worlds; an inheritance is a view whose sources live in one lineage.

Remark 6.4 (The dispatcher is the speaker)**.** *Part II [6] introduced the dispatcher $\mathcal{D} : \mathcal{B} \times \mathrm{State} \to \Delta(\mathrm{Hosts})$ as the mechanism that selects which host actualizes the shared langue. In the geographical reading, the dispatcher is the individual speaker who chooses, at each utterance, which world to speak: the learned policy over hosts is the executive side of the linguistic community's geography. Langue never speaks; worlds never speak; the dispatcher speaks, and its choices are the daily traffic in which waves travel.*

Saussure (Part Four)	In the *Course*	In HLLSet Algebra
diversity of languages	languages differ in space	the family of worlds Ω over one plane: Reading 3.1, Proposition 3.3
language vs dialect	difference of degree	one datatype (world); only LUT size, generation, and threshold differ
geographical complexity	dialect continua, isoglosses	graded overlap and threshold level sets: Definition 4.2, Proposition 4.3
cause of diversity	time, not space	unsynchronized pivots; divergence is the product of transports: Proposition 5.2
substratum	an older layer under the present	old generations readable in the DAG: Observation 5.3
linguistic wave	innovation spreading by communication	the bridge Φ: Definition 2.5, Proposition 6.2
barrier to a wave	geographical/social obstacle	missing codebook or threshold rejection: Reading 6.1
borrowing vs inheritance	loans and native forms	provenance crossing two worlds vs one lineage: Observation 6.3
the speaker	who actualizes langue	the dispatcher $\mathcal{D}$ over hosts: Part II [6], Remark on the dispatcher

Table 1: The complete dictionary of Part Four of the *Course* into HLLSet Algebra.

7 The dictionary of Part Four

8 What Part IV adds to Parts I–III

1. **The world layer becomes a linguistic object.** The EWM multi-world machinery [9] was built for isolation and bridges; Part IV reads it as Saussure's geography: worlds are languages, LUTs are lexicons, thresholds are community boundaries.
2. **Diversity is costless at the representation level.** Proposition 3.3 is the exact form of the structuralist claim that all languages are one system seen differently.
3. **The cause of diversity is time, computed.** Proposition 5.2 turns Saussure's "time is the primary cause" into a product of change-of-basis matrices.
4. **Loans and inheritances are decidable.** Observation 6.3 separates borrowed and inherited nodes by provenance, without a dictionary.
5. **The dispatcher finds its place.** The executive question of Part II becomes the geography of the speaker.

9 Conclusion and outlook to Part V

Part Four of the *Course* has been re-founded as the multi-world theory of one lattice. Diversity is a family of worlds over one plane; complexity is graded overlap with threshold boundaries; the cause of diversity is time, made exact as unsynchronized pivots; waves are bridges; substrata are old generations; loans are crossing provenances. Geographical linguistics, read in this direction, is no longer a catalogue of dialects: it is the theory of which worlds share which frames, and how innovations cross between them.

What Part Four leaves to Part Five is the direction of reading itself. We have read the DAG forward: pivots, transports, waves. Part Five of the *Course*, Questions of Retrospective

Linguistics, reads it backward: the two perspectives of diachronic linguistics, the search for the earliest language and the prototype, reconstructions, the contribution of linguistic evidence to anthropology and prehistory, and the distinction of language families from linguistic types. In the algebra these become time travel through old frames: reading the present against the first recorded basis, with the spill as the error of the journey; reconstruction as a view whose provenance points into the past; and the family/type distinction as ancestry in the DAG versus similarity in the current fiber. Part V will close the series, and with it the circle: the *Course* interpreted by the algebra, and the algebra interpreted by the *Course*.

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